



GIA - IAA

Introducció a

l'Aprenentatge Automàtic

5. Lab: Classification with Logistic Regression, KNN, SVM and Decision Trees

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Pre-processing

❖ Standarization

```
1 from sklearn.preprocessing import StandardScaler
2
3 sc = StandardScaler()
4 sc.fit(X_train)
5 X_train_std = sc.transform(X_train)
6 X_test_std = sc.transform(X_test)
```

[5] ✓ 0.0s

❖ Dimensionality reduction PCA

```
1 pca = PCA().fit(X_train);
2 X_train_pca = pca.transform(X_train)
```

[44] ✓ 0.1s

Feature Scaling is NOT Always Necessary

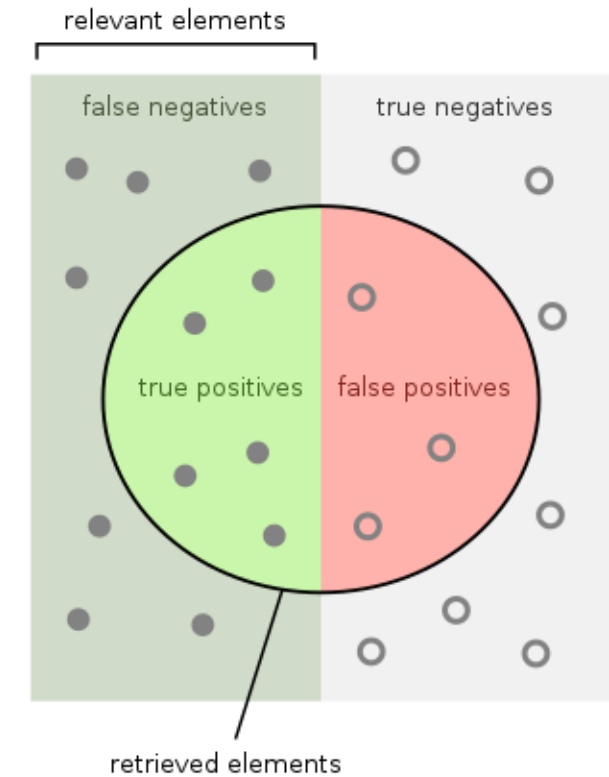
Algorithm	Test Set Classification Performance		Scaling Required?
	Without Feature Scaling	With Feature Scaling	
Logistic Regression	0.53	0.70	YES
Support Vector Classifier	0.72	0.94	YES
MLP Classifier	0.73	0.89	YES
kNN Classifier	0.66	0.93	YES
Decision Tree	0.83	0.83	NO
Random Forest	0.91	0.91	NO
Gradient Boosting	0.86	0.86	NO
Naive Bayes	0.75	0.75	NO

Better performance with feature scaling

Same performance irrespective of feature scaling

Performance Metrics

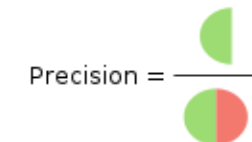
		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Total population = P + N		
	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection



$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2tp}{2tp + fp + fn}.$$

https://en.wikipedia.org/wiki/Precision_and_recall

How many retrieved items are relevant?



How many relevant items are retrieved?

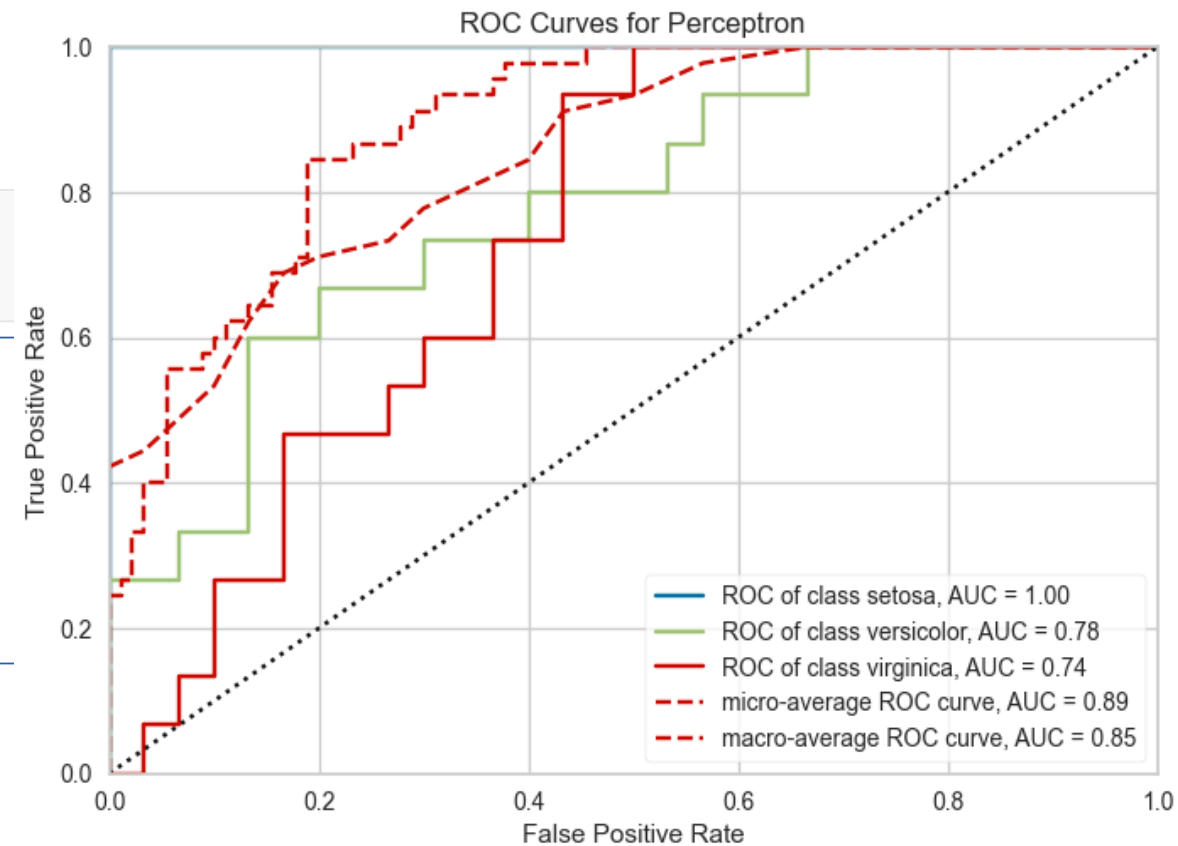


Performance metrics

```
1 print(classification_report(ppn.predict(X_test_std), y_test, target_names=iris.target_names))
2 results_df = save_results(ppn, X_test_std, y_test, 'Perceptron', results_df)
```

✓ 0.0s

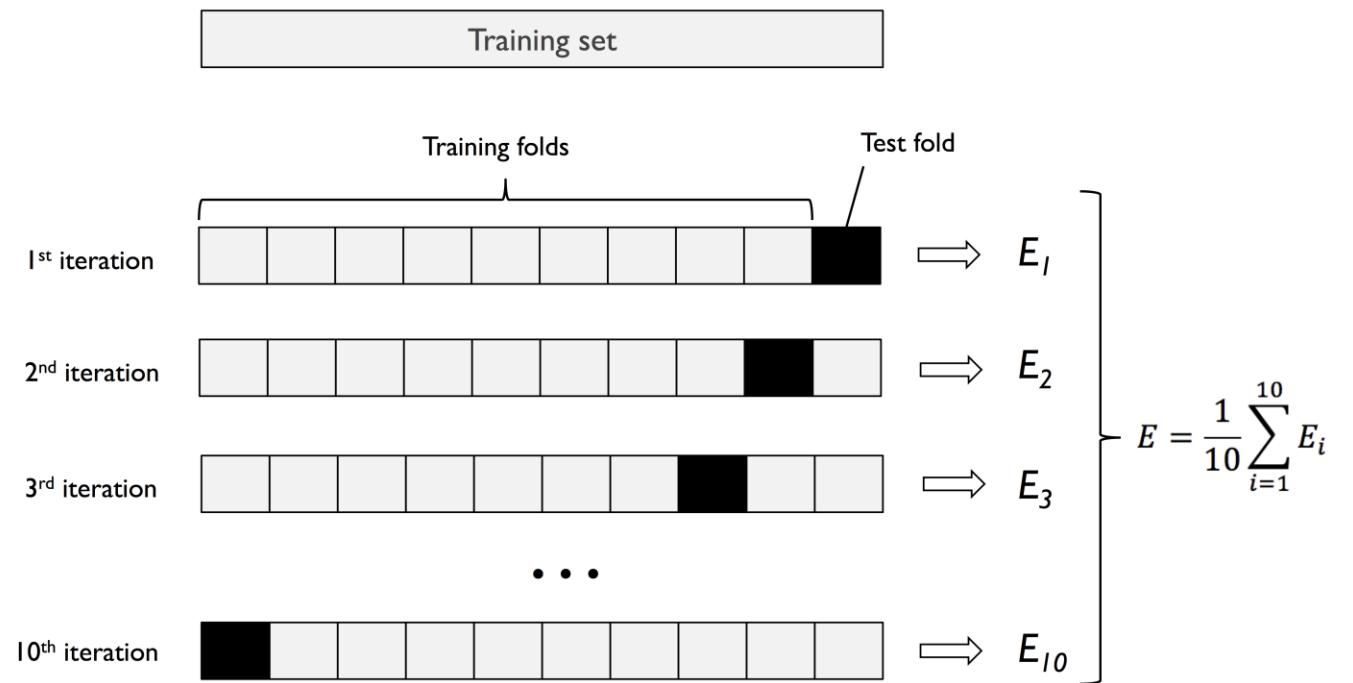
	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	15
versicolor	0.13	0.67	0.22	3
virginica	0.93	0.52	0.67	27
accuracy			0.69	45
macro avg	0.69	0.73	0.63	45
weighted avg	0.90	0.69	0.75	45



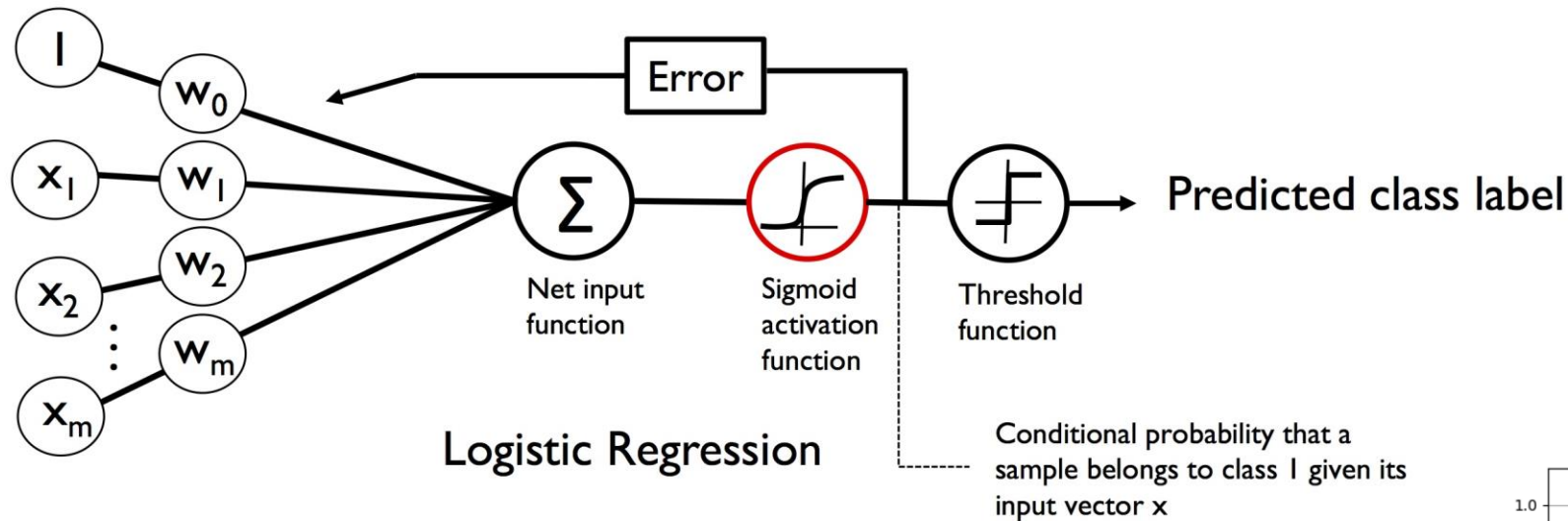
Receiver Operating Characteristic curve

K-fold cross validation

```
1 print(np.mean(cross_val_score(lr,X_train_std,y_train,cv=10)))  
[463] ✓ 0.0s  
... 0.8772727272727273
```



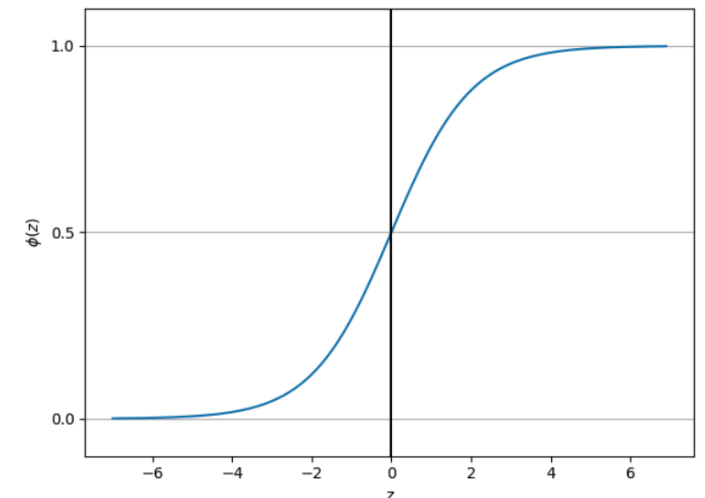
Logistic Regression



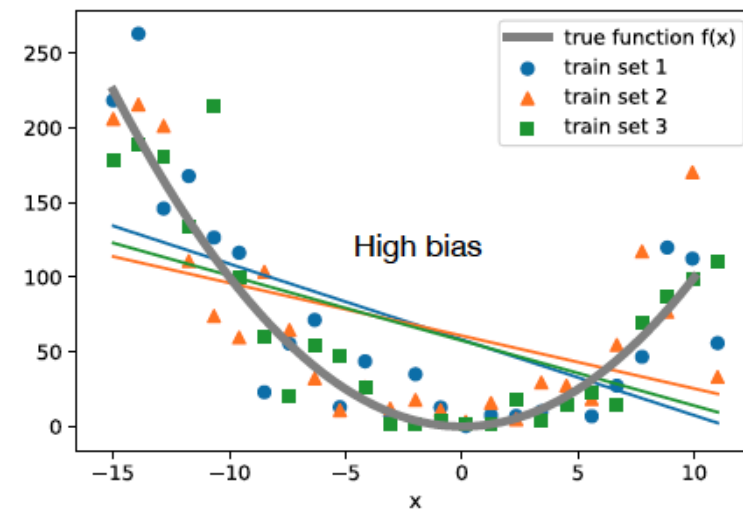
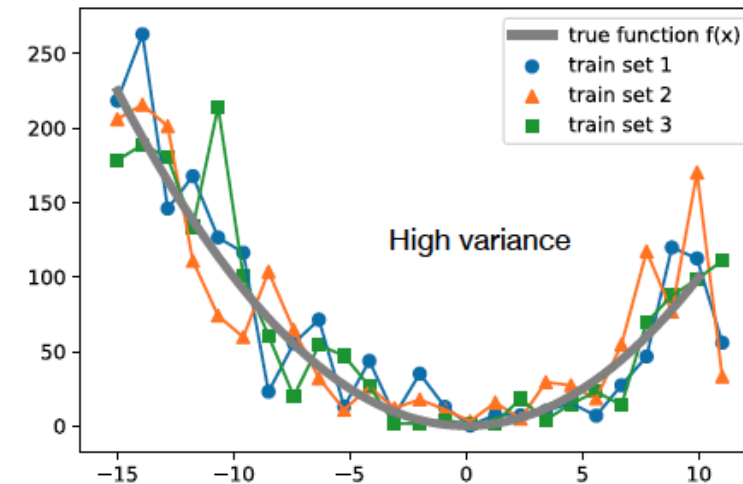
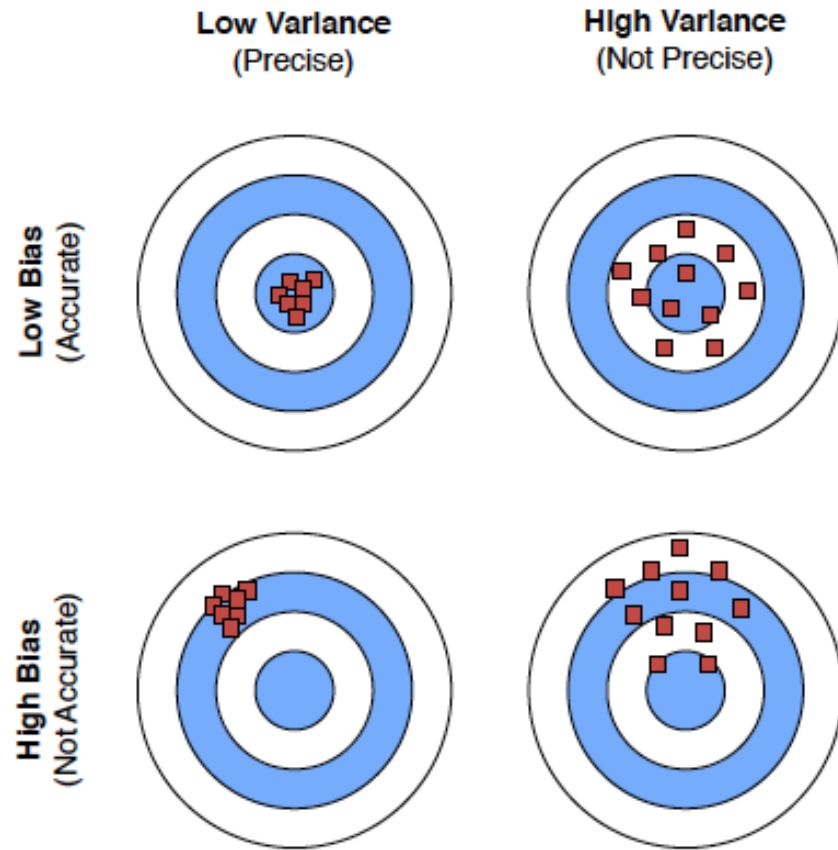
$$\phi(z) = \frac{1}{1 + e^{-z}}$$

Here, z is the net input, the linear combination of weights, and the inputs (that is, the features associated with the training examples):

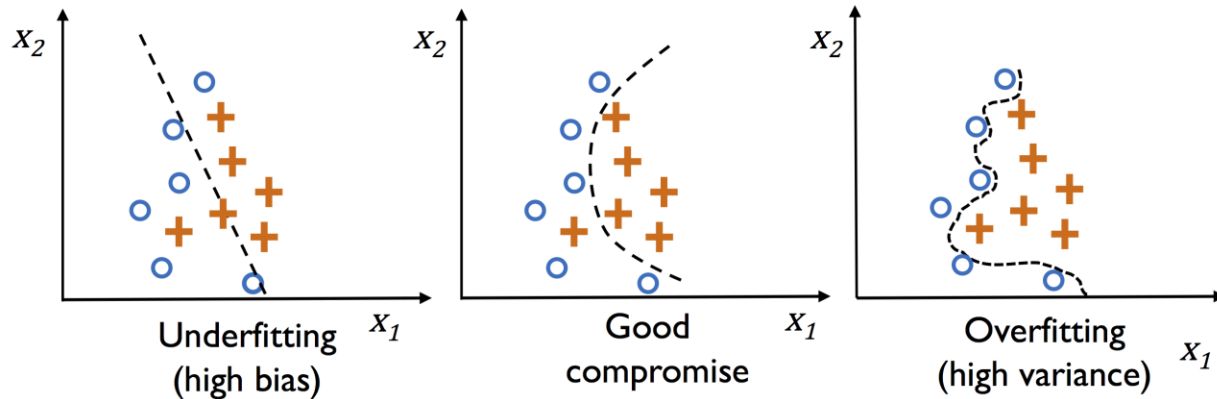
$$z = \mathbf{w}^T \mathbf{x} = w_0 x_0 + w_1 x_1 + \dots + w_m x_m$$



Overfitting and Underfitting: High Variance and High Bias



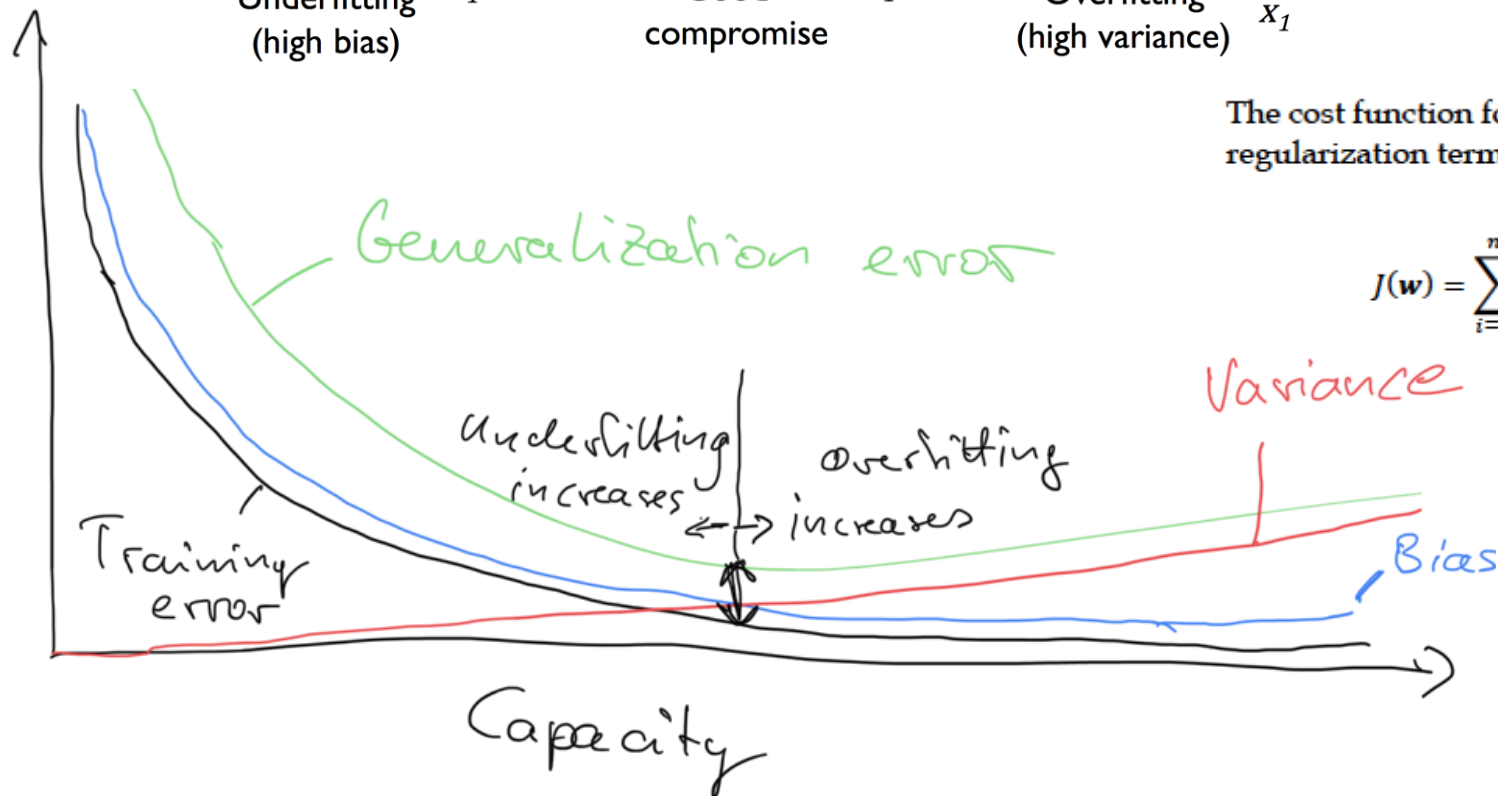
Summary



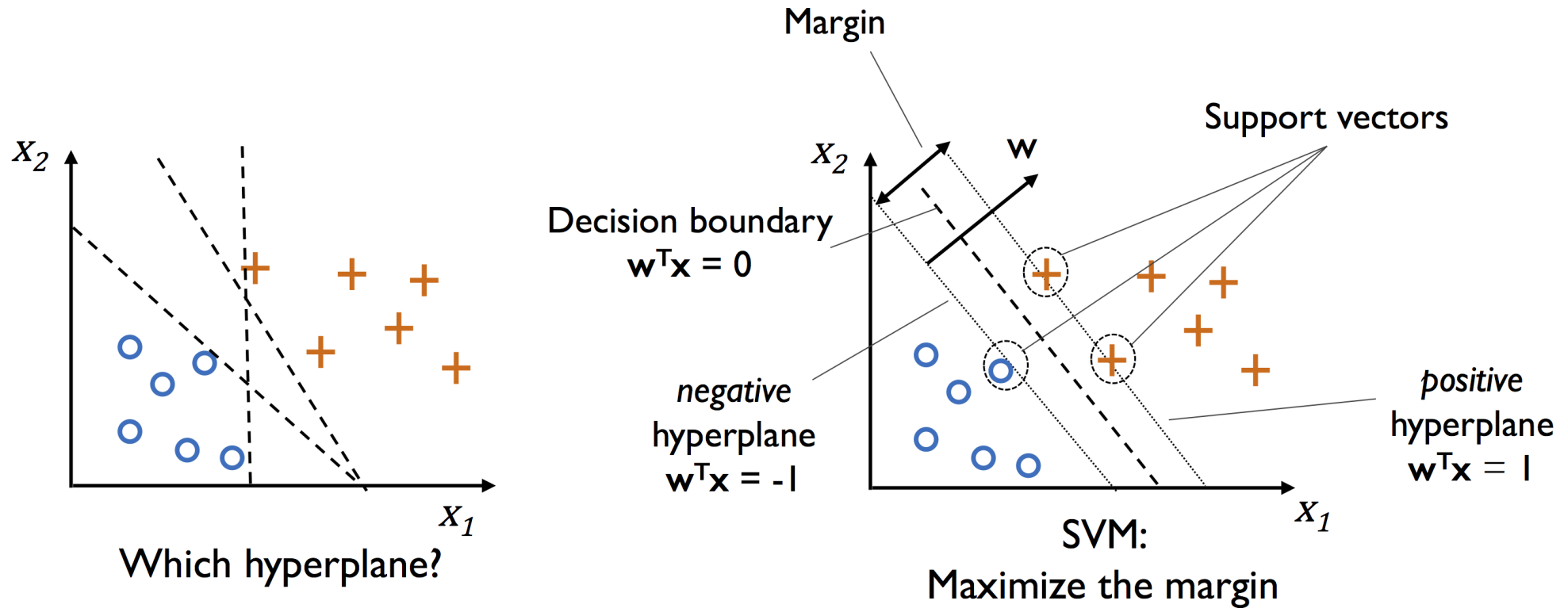
❖ What's is λ and how it is related to C in LogisticRegression model?

The cost function for logistic regression can be regularized by adding a simple regularization term, which will shrink the weights during model training:

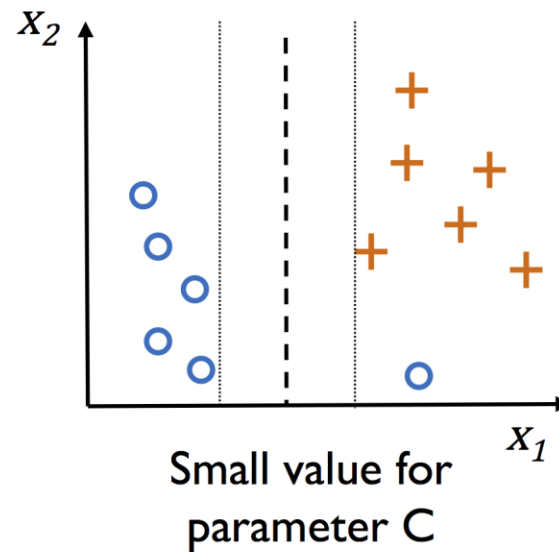
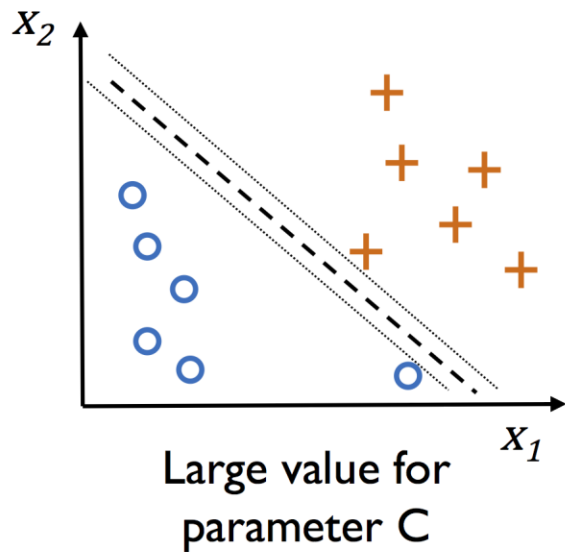
$$J(\mathbf{w}) = \sum_{i=1}^n \left[-y^{(i)} \log(\phi(z^{(i)})) - (1 - y^{(i)}) \log(1 - \phi(z^{(i)})) \right] + \frac{\lambda}{2} \|\mathbf{w}\|^2$$



Support Vector Machines



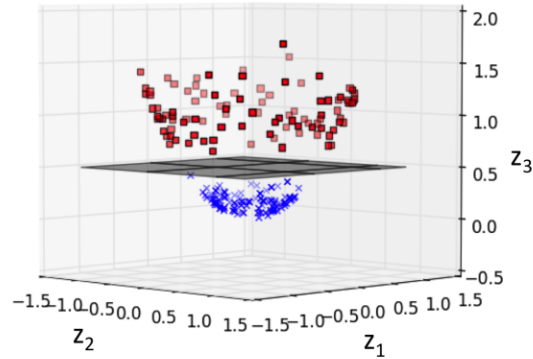
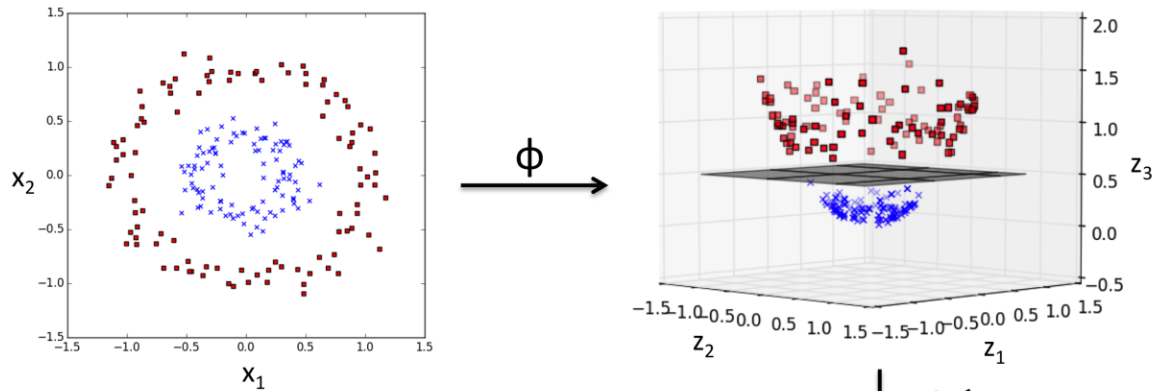
Support Vector Machines



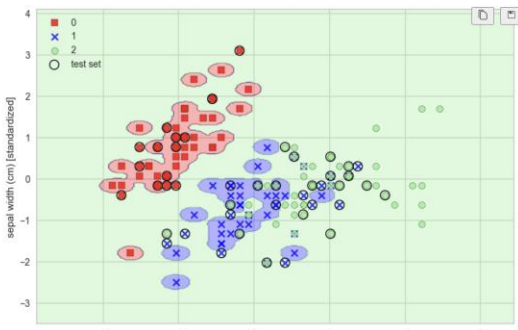
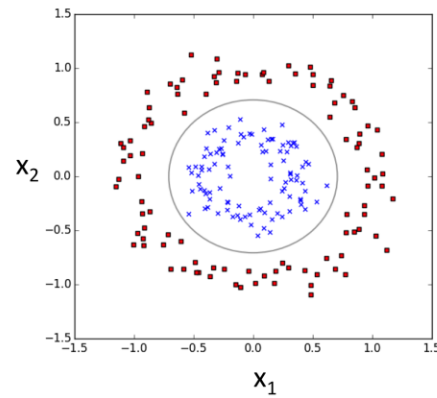
$$\frac{1}{2} \|w\|^2 + C \left(\sum_i \xi^{(i)} \right)$$

- ❖ Vary the penalty control parameter C in SVM model to control the width of the margin between classes.

Solving non-linear problems using a kernel SVM



ϕ^{-1}



- ❖ Play with the kernels in SVM scikit model to solve the XOR problem. What's gamma? How it affects the boundaries between classes? How it relates to bias and variance of the model?

One of the most widely used kernels is the **radial basis function (RBF) kernel**, which can simply be called the **Gaussian kernel**:

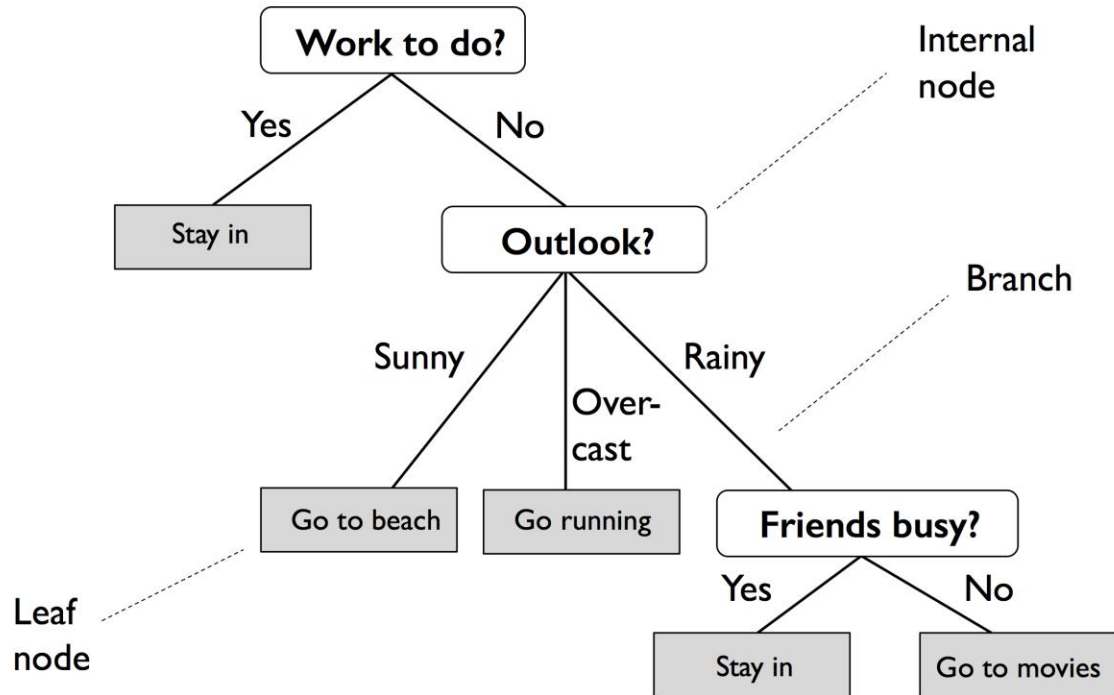
$$\kappa(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = \exp\left(-\frac{\|\mathbf{x}^{(i)} - \mathbf{x}^{(j)}\|^2}{2\sigma^2}\right)$$

This is often simplified to:

$$\kappa(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = \exp\left(-\gamma\|\mathbf{x}^{(i)} - \mathbf{x}^{(j)}\|^2\right)$$

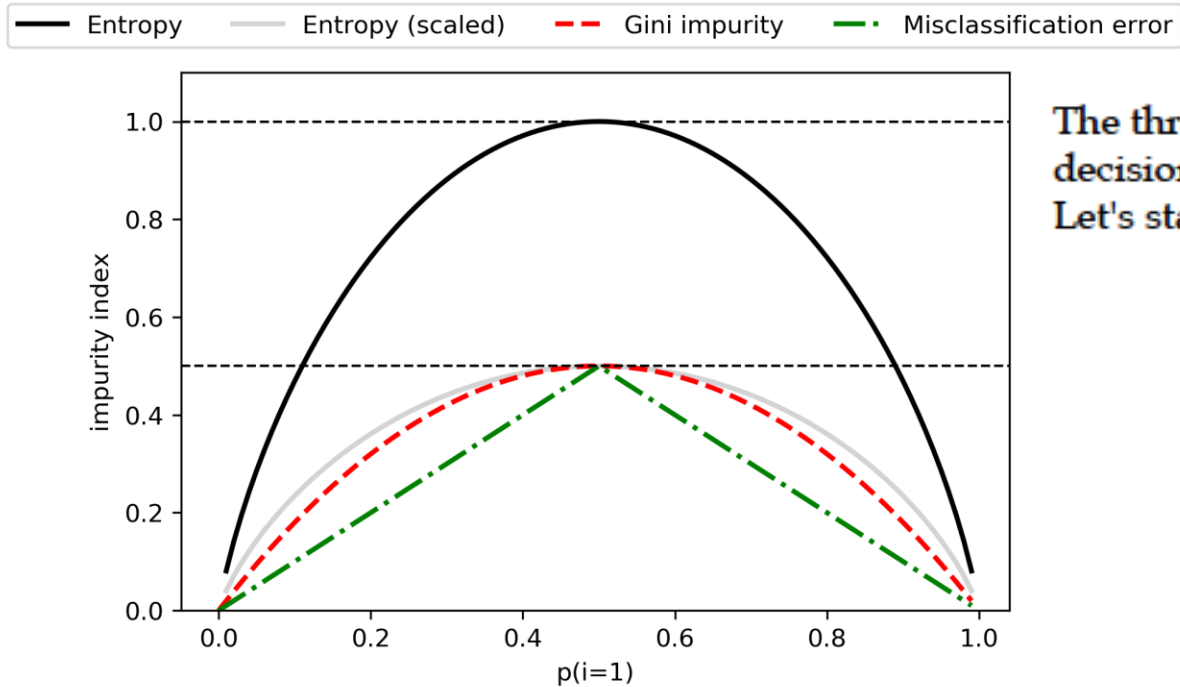
Here, $\gamma = \frac{1}{2\sigma^2}$ is a free parameter to be optimized.

Decision Trees



- ❖ For a continuous real numbers, we can ask also questions simply defining a cut-off: Is the sepal width $\geq 1.5\text{cm}$?

Decision Trees



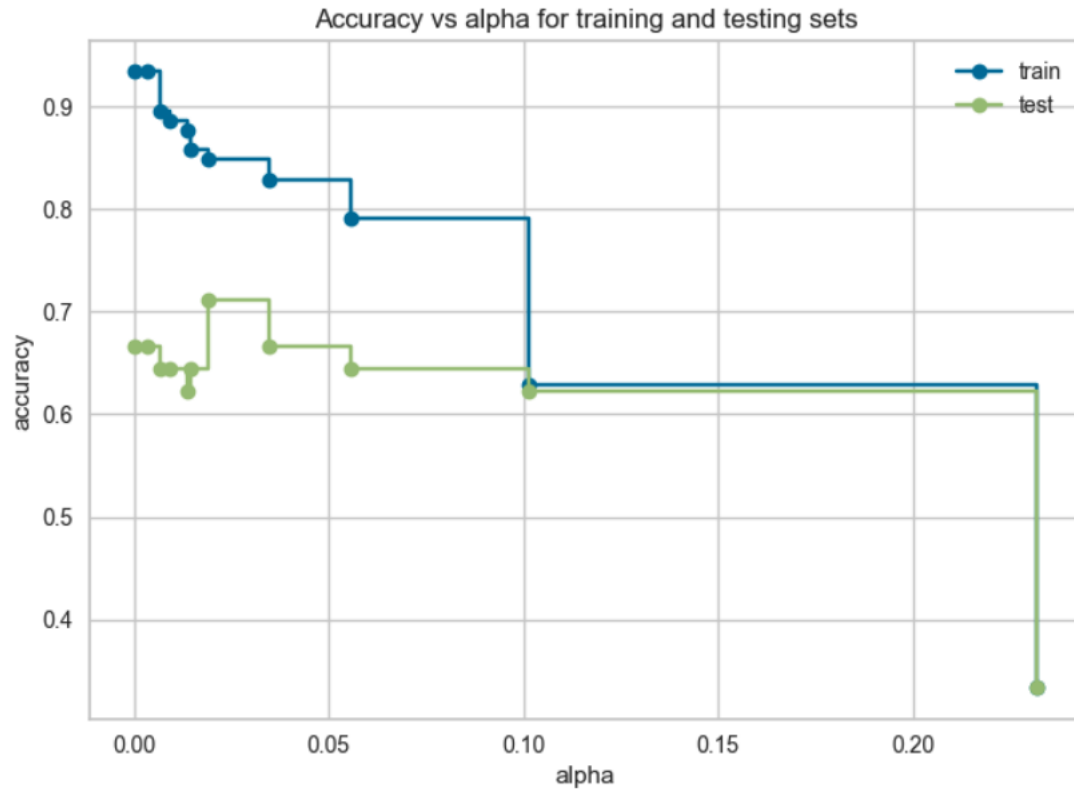
The three impurity measures or splitting criteria that are commonly used in binary decision trees are Gini impurity (I_G), entropy (I_H), and the classification error (I_E). Let's start with the definition of entropy for all non-empty classes ($p(i|t) \neq 0$):

$$I_H(t) = - \sum_{i=1}^c p(i|t) \log_2 p(i|t)$$

$$I_G(t) = \sum_{i=1}^c p(i|t)(1 - p(i|t)) = 1 - \sum_{i=1}^c p(i|t)^2$$

$$I_E(t) = 1 - \max\{p(i|t)\}$$

Decision Trees

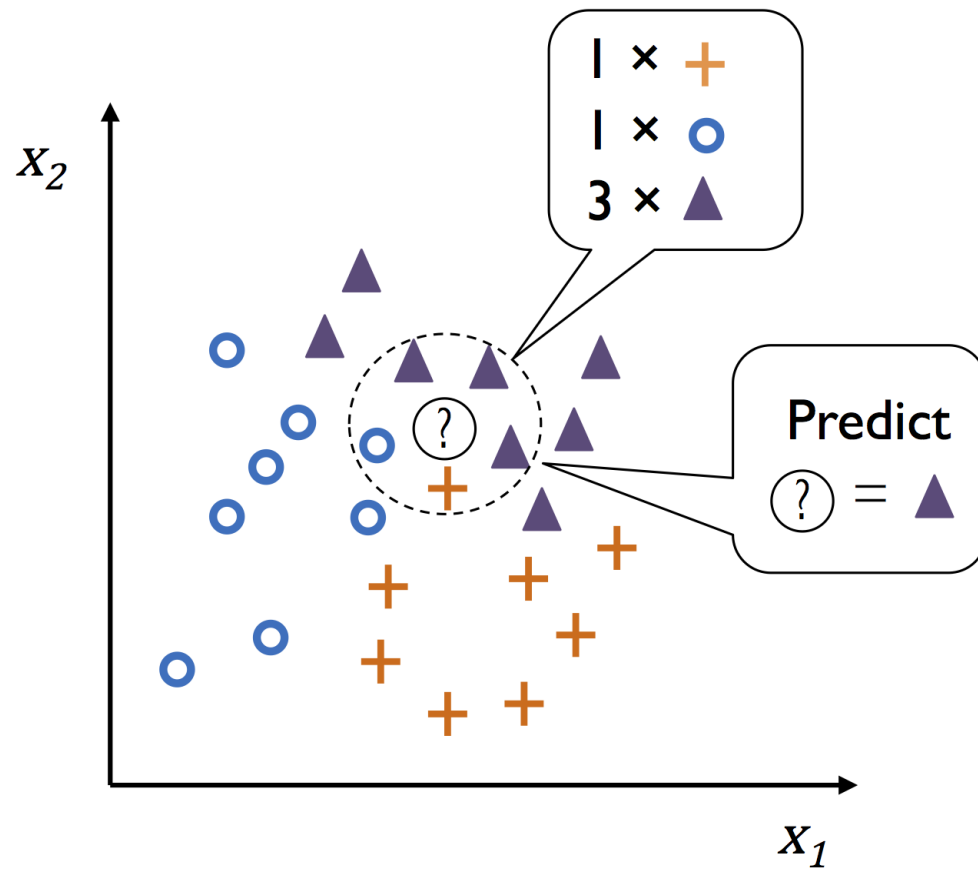


- ❖ Play with alpha parameter and plot a graph for Accuracy VS alpha for the Iris dataset. Use the method `score()` to access the predict method and metrics from the models.



```
1 train_scores = [tree_model.score(X_train, y_train) for tree_model in tree_models]
2 test_scores = [tree_model.score(X_test, y_test) for tree_model in tree_models]
3
```

K-Nearest Neighbors



- ❖ Play with the number of k-neighbors and the metric. Perform a grid-search varying such parameters.